



Cambridge IGCSE™

MATHEMATICS

0580/22

Paper 2 (Extended)

February/March 2022

MARK SCHEME

Maximum Mark: 70

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **8** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfw	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1	40°	1	
2	80.50 cao	2	B1 for 80.498... or 80.5 or correctly rounding their more accurate decimal to 2 dp
3	7 [h] 18 [min]	1	
4	166	3	M2 for $[2 \times] (7 \times 4 + 4 \times 5 + 5 \times 7)$ or M1 for 7×4 or 4×5 or 5×7
5(a)	5	1	
5(b)	(0, 7)	1	
6	correct triangle with arcs	2	B1 for correct triangle with incorrect or no arcs or for two correct arcs. or a triangle with arcs but one side not in range
7	$n > -1$ oe	1	
8(a)(i)	triangle at $(-1, 1)$ $(-4, 2)$ $(-3, 5)$	1	
8(a)(ii)	triangle at $(-2, -3)$ $(1, -2)$ $(0, 1)$	2	B1 for translation by $\begin{pmatrix} -3 \\ k \end{pmatrix}$ or by $\begin{pmatrix} k \\ -4 \end{pmatrix}$
8(b)	enlargement [sf] $\frac{1}{2}$ [centre] $(9, -1)$	3	B1 for each
9	$3a(4a^2 - 7)$ final answer	2	B1 for $3(4a^3 - 7a)$ or $a(12a^2 - 21)$ or for $3a(4a^2 - 7)$ seen then spoilt
10(a)	8 11 16	2	B1 for two correct
10(b)	$23 - 8n$ oe final answer	2	B1 for $j - 8n$ or $23 - kn$ $k \neq 0$ or $23 - 8n$ seen then spoilt
11	Positive	1	

Question	Answer	Marks	Partial Marks
12(a)	805	3	B2 for 105 or M2 for $\frac{700 \times 2.5 \times 6}{100} + 700$ oe or M1 for $\frac{700 \times 2.5 [\times 6]}{100}$ oe
12(b)	2.3[0...]	3	M2 for $\sqrt[17]{\frac{1030.35}{700}}$ oe or M1 for $1030.35 = 700(k)^{17}$ oe for any k
13(a)	h^7 final answer	1	
13(b)	$\frac{x^3}{343}$ final answer	1	
13(c)	6	1	
14	29.5 or 29.53...	2	M1 for $2 \times \pi \times 4.7$ oe
15	$\frac{7}{3}$ oe improper fraction	M1	or $\frac{k}{3} \times \frac{11}{14}$ where $k > 3$
	$1\frac{5}{6}$ cao	A2	A1 for $\frac{77}{42}$ or $\frac{11}{6}$ or $1\frac{35}{42}$
16(a)	$[y =] -2x - 7$ final answer	2	B1 for $-2x + c$ or $kx - 7$, $k \neq 0$ final answer
16(b)	$y = \frac{1}{2}x[\pm 0]$ final answer	2	FT $-\frac{1}{\text{their gradient in (a)}}$ B1 for $y = kx[\pm 0]$ oe, $k \neq 0$ or $y = \text{their} \frac{1}{2}x + c$ oe for any c or $\text{their} \frac{1}{2}x [\pm 0]$ oe

Question	Answer	Marks	Partial Marks
17	77.8 or 77.77 to 77.80	5	B4 for answer 22.2[%] or 22.20[%] to 22.23[%] OR M1 for $\tan^{-1} \frac{11}{4}$ oe or $\tan^{-1} \frac{4}{11}$ oe M2 for $4 \times 11 - \frac{\text{their acute angle}}{360} \times \pi \times 4^2$ oe or M1 for $\frac{\text{their acute angle}}{360} \pi \times 4^2$ oe M1 for $\frac{\text{their shaded area}}{4 \times 11} [\times 100]$ oe or $\frac{\text{their sector area}}{4 \times 11} \times 100$ oe
18	A correct equation leading to 41	3	M2 for $4x = 164$ or M1 for $x + 2(x - 24) + x - 16 = 100$ oe or M1 for correctly simplifying <i>their</i> equation to the form $kx = c$ provided at least one part correct from $[2](x - 24)$ oe or $x - 16$ or B1 for answer 41 without an equation in x shown
19	$\frac{2}{3}$ oe	3	M1 for $y = \frac{k}{\sqrt{x+4}}$ M1 for $y = \frac{\text{their } k}{\sqrt{77+4}}$

Question	Answer	Marks	Partial Marks
20	$x^2 + 6x - 40$ [=0] or $y^2 - 40y - 41$ [=0]	M2	M1 for correct method to eliminate one variable e.g. $x^2 - 2(11 - 3x) = 18$ or $\frac{(11 - y)^2}{3^2} - 2y = 18$
	$(x - 4)(x + 10)$ [=0] or $(y - 41)(y + 1)$ [=0]	M1	or for correct factors for <i>their</i> quadratic equation or for correct use of quadratic formula for <i>their</i> quadratic equation or for correctly completing the square for <i>their</i> quadratic equation
	$x = 4, y = -1$ $x = -10, y = 41$	B2	B1 for $x = 4, x = -10$ or for $y = -1, y = 41$ or for a correct pair of x and y values If B0 scored and at least 1 method mark scored SC1 for correct substitution shown of both of <i>their</i> x values or <i>their</i> y values into $3x + y = 11$ or $x^2 - 2y = 18$
21(a)	35.1 or 35.05 to 35.06	4	M3 for $\tan = \frac{14.5}{\sqrt{18.6^2 + 9^2}}$ oe or M2 for $[AC^2 =]18.6^2 + 9^2$ oe or better or $[AG^2 =]18.6^2 + 9^2 + 14.5^2$ or M1 for recognising the angle GAC

Question	Answer	Marks	Partial Marks
21(b)	$30 - \sqrt{18.6^2 + 9^2 + 14.5^2}$ $30 - \frac{14.5}{\sin(\text{their}(\mathbf{a}))}$ or $30 - \frac{\sqrt{18.6^2 + 9^2}}{\cos(\text{their}(\mathbf{a}))}$	M2	M1 for $AG^2 = 18.6^2 + 9^2 + 14.5^2$ oe or better or $\sin(\text{their}(\mathbf{a})) = \frac{14.5}{AG}$ or $\cos(\text{their}(\mathbf{a})) = \frac{\sqrt{18.6^2 + 9^2}}{AG}$
	4.75 to 4.78...	A1	
22(a)	$a - \frac{2}{5}b$ oe simplified	2	M1 for $-b + a + \frac{3}{5}b$ or a correct route
22(b)	$\frac{5}{2}a$ oe	2	B1 for ka where $k > 1$ or $\frac{5}{2}\overline{OR}$